Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/07/2016
PI Name:	Weil, Michael Ph.D.		
Project Title:	Characterization of the Tumor Spectrum Arising in HZE Ion Irradiated Outbred Mice		
Division Name:	Human Research		
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Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	80521-2807	**Congressional District:**	4
Comments:			
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No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Assignment:	NOTE: End date change to 5/09/2016 per S. Monk/LaRC/HRP (Ed., 10/14/15)		
Key Personnel Changes/Previous PI:	No changes in year 2.		
COI Name (Institution):	Belknap, John (Oregon Health & Science University) Raber, Jacob (Oregon Health and Science University) Hall, Eric (Columbia University) Kleiman, Norman (Columbia University)		
Grant/Contract No.:	NNX12AB54G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal has two key objectives. The first is to compare the spectrum of tumor types that arises in a genetically diverse mouse population irradiated with HZE (high energy) ions to the spectra arising in similar unirradiated or gamma-ray irradiated populations. The second objective is to identify genetic differences between individual outbred mice that make them susceptible or resistant to the induction of specific types of tumors by radiation and then to determine if the same genetic factors that control sensitivity to gamma-ray induced tumors also control sensitivity to HZE ion induced tumors. To accomplish these objectives we will irradiate mice from an outbred, heterogeneous stock, HS/NPT, with HZE ions or gamma-rays and monitor them along with unirradiated control mice for tumor development. Tumors that arise in the mice will be classified by histological examination and the tumor spectra will be compared between the three groups. We will identify the genetic differences, known as QTL or quantitative trait loci, underlying susceptibility to HZE ion-induced and gamma-ray-induced tumorigenesis and determine their overlap by analyzing single nucleotide polymorphisms (SNP) throughout the genomes of tumor bearing and unaffected mice for association with specific tumor types. The proposed work addresses the solicitation's emphasis on studies of the individual basis of space radiation risk assessment including the "use of genetically diverse populations of mice to represent human heterogeneity that will elucidate the spectrum of tumor types caused by HZE nuclei as well as possible differences compared to gamma-rays." In addition, it addresses NASA identified knowledge gaps related to uncertainties in risk projections arising from individual radiation sensitivity and gender. Finally, the proposed study has the potential to identify experimental models for tumor development in tissues for which such models are lacking or limited such as colon, stomach, bladder, skin, esophagus, and brain.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Radiation carcinogenesis is a matter of concern for individuals exposed to radiation for medical diagnosis or treatment, for occupational reasons, or because of accident. This study will provide basic information about genetic susceptibility to radiation-induced cancers.

SUPPLEMENTAL INFORMATION FOR FINAL REPORT (Submitted September 2016)
The focus during the current report year was on completing the remaining analyses and the preparation and submission of the corresponding manuscripts.

Carcinogenesis.

To determine whether tumors that arise following HZE ion exposure are more malignant than their counterparts arising in unirradiated or gamma-ray irradiated mice, we characterized two measures of malignancy: tumor latencies (analyzed by Kaplan-Meir survival procedures) and metastatic events. As expected, tumors arising in both HZE ion and gamma-ray irradiated mice showed significantly decreased latencies. However, HZE ions did not produce tumors with decreased latencies when compared to tumors in gamma-ray irradiated mice. Pulmonary metastases were observed in cases of hepatocellular carcinoma, Harderian gland adenocarcinoma, osteosarcoma, and ovarian granulosa cell tumor. Metastases were no more frequent in irradiated animals than in controls and there was no significant difference in metastatic incidence between HZE ion irradiated mice and gamma-ray irradiated mice.

The DOQTL program, which is perhaps the most advanced program available for QTL analysis in heterogeneous stock mice, was re-written so it could be used with genotyping data from HS/Npt mice and categorical and time to event traits, the trait types most relevant for cancer and cataract studies. Sixteen neoplastic subtypes occurred with a sufficient sample size (greater than 1% incidence or a minimum of 6 cases) to perform comparisons of QTL results between radiation exposure groups. In total, 114 QTL were mapped for the 16 neoplasms with an average 95% confidence interval of 2.7 Mb. In some cases candidate genes for control of susceptibility to specific cancers could be identified.

To detect possible coincident QTL between tumor types and exposure types using a holistic genome-wide method, rather than using individual selected QTL regions, genome-wide association scans were utilized in a clustering procedure to determine coincident loci. The results of this analysis indicated that tumor types often clustered together regardless of radiation exposure. This finding further supports the hypothesis that host genetic factors are more important in determining neoplasm incidence than radiation exposure type.

Ocular effects.

The cataract data was analyzed in conjunction with the SNP genotyping data to identify quantitative trait loci (QTL) that control susceptibility to radiogenic cataracts. The mapping phenotype was days to grade 2 cataract and genome wide scans were performed using the DOQTL program which we have modified for use with Cox regression. Significance at 80% and 95% confidence was determined by permutation. Several QTL were identified at 95% confidence and one, on chromosome 13, was narrow enough that a strong candidate gene emerged. A manuscript has been prepared for submission.

Cognitive effects data analysis.

As described earlier, about 1,850 mice were irradiated and behaviorally tested in two cohorts of 600 and 1,200 mice, respectively. In our statistical analysis for behavioral and genetic analyses, we detected a significant cohort effect. This might be due to the fact that urine was collected 24 hours following sham-irradiation or irradiation for NMR analysis as part of the second cohort but this additional procedure was not part of the experience of the first cohort of mice. We assessed whether the NMR data was related to or could predict behavioral or cognitive outcome measures three months later but we did not detect evidence for that. We also did not detect an effect of irradiation on the NMR data. Based on the cohort effect described earlier, we analyzed the data separately and focused the main analysis on first cohort of about 600 mice.

Our genetic data consists of ~77K genetic markers across the genome. Among these, about 40K markers were deemed as informative, i.e., they had a minor allele frequency above 5%. As described below, the utilization of this set of informative markers increased the robustness and reproducibility of the results. However, we recognize that the low frequency alleles could also be informative in some cases and/or have a strong effect on some phenotypes.

The QTL analysis is based on more than 40 behavioral and cognitive phenotypes for radiobiological outcome measures of interest related to performance in the hippocampus-dependent contextual and hippocampus-independent cued fear conditioning tasks. This large collection of related phenotypes and genotypes required the adoption of a data reduction technique that 1) summarizes the essential features of the behavioral and cognitive measures and 2) enhances the ability to detect significant genotype-phenotype associations.

For the behavioral and cognitive measurements, based on the observation that many of the measurements display significant

Task Progress:

correlations, we employed an analysis strategy based on a Principal Component Analysis (PCA). Our results show that the first 5 principal components (PCs) contain more than 65% of the variability of the data and each of the remaining individual PCs contain little additional variance. As predicated by the PCA, all the principal components are linearly independent. Moreover, by examining the "loadings" or the projections of the original behavioral and cognitive measures on the PCs, it is apparent that related behavioral and cognitive phenotypes map onto distinct PCs. This observation verifies that of the original. Additionally, it appears that distinct behavioral and cognitive domains map into distinct principal components, as quantified by the "loadings," or the level of association between the original ~40 behavioral measurements and the reduced 5 PCs. What is striking is that for example PC1 contains cognitive measures of hippocampus-dependent fear memory (contextual fear memory, i.e., a fear memory in the same environment or context as where the aversive stimulus was experienced during training), while PC3 contains cognitive measures of hippocampus-independent fear memory (cued fear memory, i.e., fear memory in a novel environment). PC2 contains cognitive measures in response to the aversive stimulus during training, while PC5 contains cognitive measures of performance at baseline prior to training (prior to an aversive stimulus) and during the training (i.e., learning). Finally, PC4 contains cognitive measures) during training (i.e., learning).

Our data illustrate that the behavioral and cognitive phenotypes can efficiently be represented by a much smaller set of principal components.

The second major component of our analysis is to relate the behavioral and cognitive phenotypes, as represented by the PCs, to the genotype data. For this, we combined two analysis methods. First, we used multivariate distance matrix regression (MDMR), which relates P variables to M factors collected on N individuals, where P>>N. This analysis method can be applied in principle to any data modality, provided that a suitable similarity measure can be devised for each type of data. MDMR analysis involves computing the distance between all pairs of individuals with respect to P variables of interest and constructing an N×N matrix whose elements reflect these distances. Permutation tests are used to test hypotheses that consider whether or not the M additional factors collected on the individuals can explain variation in the observed distances between and among the N individuals as reflected in the matrix.

An essential step in the MDMR procedure involves selecting appropriate distance measures for each data modality, which include single nucleotide polymorphisms (SNPs) for the genotype and PCs for the behavioral and cognitive data. Both the genotype data and the behavioral and cognitive summary measurements are represented as a set of pairwise distances. For the genotype data, we utilized the Manhattan distance, which assigns an equal weight to each genotype. For the behavioral and cognitive measurements, we used the absolute value difference as a distance measure.

The MDMR procedure is complemented by a second analysis step based on machine learning principles. Once all the marker-phenotype correlations are computed, the resulting values are ranked in decreasing order. If and only if an individual association reaches statistical significance, we proceed to the next step, which involves selecting a subset of the M factors that are more informative than either the most informative single factor, or the combination of all factors. A new composite genotype distance measure is evaluated and new markers/intervals are added to the analysis. The analysis is stopped when the combined correlation reaches a peak value. This procedure is denoted in the machine learning literature as "forward selection." We randomly subsampled the N individuals and recorded the number of times that each genetic marker/interval got selected as among the most informative. The final set of factors includes only those detected as informative in a high percentage (>75%) of the bootstrapped models. The analysis capturing the Mantel correlation for each genetic marker with PC1 shows that the vast majority of the ~40K markers are not associated with the PC1 "phenotype," as expected. However, there is small/moderate association for 100-200 markers. Moreover, when markers are combined and evaluated together, the cumulative correlation with the phenotype achieves much greater values. This step in our analysis departs from a conventional QTL analysis, which often focuses on the few markers that achieve the highest significance and survive multiple testing adjustments. However, this procedure can severely underestimate the total genetic component of the phenotype variability. Our procedure focuses on a set of markers that taken together explain a larger portion of the phenotypic variability, even though each individual marker might not achieve high statistical significance. To guard against spurious results, we adopted the bootstrap aggregating procedure described above.

By utilizing the methodologies outlined above, we detected sets of informative markers for each of the five most significant principal components. This analysis was performed independently for each radiation type. We next inquired whether different principal components and different radiation types show any overlap in term of the genetic factors that influence them. In both cases, we detected very little overlap. While the principal components are independent and therefore it is expected that they will be under the control of distinct genotypes, the low level of overlap between the same phenotypes affected by different radiation types is remarkable.

These findings suggest that following distinct radiation exposure, the genotype/phenotype relationship shifts. Importantly, our bootstrap aggregating procedure reveals that some markers are robustly detected under one radiation type, but very rarely detected under a different radiation type.

The final analysis step leverages the selection of informative markers described above. Our goal is to identify a set of genetic markers that can be utilized as biomarkers for propensity for behavioral and cognitive alterations following radiation exposure. Towards this goal, we selected the animals that were at the opposite extremes in terms of behavioral and cognitive performance, which in our case were the five principal components. Next, we asked whether a novel sample can be classified as Low/High based exclusively on genetic information. The computation of pairwise genetic distances facilitates the adoption of the "K nearest neighbors" algorithm. As the name suggests, for each novel sample, the K (previously categorized) samples are identified. These K nearest neighbors "vote" on the identity of the new sample. In this manner, it is feasible to infer information about the potential behavioral and cognitive profile of a sample, following a particular radiation environment, based solely on genetic information.

Pertinent to this project in which tumor and behavioral phenotypes are being analyzed in the same mice, a medical student in the Raber Lab (Wendy McGinnis), investigates the effects of immunotherapy and radiation treatment alone and combined in both healthy and tumor-bearing mice using behavioral and cognitive testing and molecular outcome measures. For this project, as part of an active collaboration with Dr. Gough at Providence, a lung carcinoma model in C57BL/6 mice is treated with immunotherapy and peripheral hypofractionated radiation therapy. Consistent with our hypothesis, combined radiation treatment and immunotherapy negatively affects behavioral and cognitive performance and activation of microglia in the brain. A manuscript with these data is currently under review.

ANNUAL PROGRESS SEPTEMBER 2014 to NOVEMBER 2015

The focus during the current report year was on completing the remaining histopathology and organizing and analyzing the data.

Ocular effects. The records of all slit lamp exams were recorded in a spread sheet. Kaplan-Meier analyses of time to grade 2 cataracts (those that are vision impairing) were made for each radiation group. Cataractogenesis was accelerated in irradiated mice, but was similar for mice irradiated with HZE ions (0.4 Gy of 600 MeV/n 56Fe or 240 MeV/n 28Si) and 3 Gy gamma-rays. There appear to be differences in cataract susceptibility between “families” of mice, but a more in depth analysis of familial susceptibility is required and will be done in the remaining 6 months.

Carcinogenesis. The histopathological evaluations proposed in the project were completed and we started a more detailed examination of the hematopoietic malignancies, beyond what was originally proposed. This includes immunophenotyping the lymphoma cases and screening the myeloid leukemias for characteristic chromosome 2 deletions. The goal is to improve the resolution of the QTL mapping of the genomic regions underlying susceptibility to radiogenic leukemia and lymphoma.

The histopathology data has now been checked for errors and placed in a format suitable for QTL analyses with the DOQTL program. This is perhaps the most advanced program available for QTL analysis in heterogeneous stock mice, but parts of the program needed to be re-written so it could be used with genotyping data from HS/Npt mice and it could be used with categorical and time to event traits rather than continuous traits. These modifications were completed and are now being tested.

Cognitive Effects Data Analysis. As a pilot project, we analyzed incomplete data sets of 600 or 1,200 SNP genotyped mice. The analyses were on the cognitive phenotypes because the assays for those phenotypes were completed early in the study.

Of the 77.8K SNP markers tested, we identified ~40K markers with sufficient genetic diversity (minor allele frequency above 5%). The mapping resolution is in the 200 kb to 2 Mb range for this HS mapping population and all of the founder strains have been completely sequenced, so promising candidate genes are very likely to emerge from the study. The QTL analysis is based on close to 40 behavioral and cognitive phenotypes for radiobiological endpoints of interest related to performance in the hippocampus-dependent contextual and hippocampus-independent cued fear conditioning tasks. We used a custom QTL detection method to scan for genetic loci that influence behavioral phenotypes in the context of irradiation (PMID: 22969789). Briefly, we start with the most significant marker, keep adding markers to the analysis, one by one in decreasing order of significance. Next, we compute the correlation between the GROUP (based on distances and Mantel correlations, which correlate a group of markers with the phenotype). Two N by N matrices are calculated, where N is the number of animals. The first matrix captures how genetically similar any two animals are, in terms of shared alleles at the selected markers. The other matrix captures how different two animals are in terms of the phenotype, just absolute value difference. The Mantel correlation is the correlation between those two matrices. Eventually we reach a peak correlation and adding any more markers starts to decrease the correlation. We select the markers with best performance and we try to classify the mice based on these markers.

As indicated earlier, two cohorts of about 600 and 1,200 mice each were sham-irradiated or irradiated with 0.4 Gy 56Fe (600 MeV.n), 0.4 Gy 28Si (240 MeV/n), or 3 Gy gamma-rays. Three months later, the mice were tested for contextual (hippocampus-dependent) and cued (amygdala-dependent) fear conditioning. Our data analysis showed that the genotype-phenotype relationships are affected by radiation type and environment (cohort). The mean and variance of a majority of phenotypes differ in the two groups of animals. Potential reasons for this gene x environment interaction is that the 600 mice were the first litter of the breeder mice of the 50 families while the 1,200 mice were the second litter, the longer time window required to test 1,200 than 600 mice increasing the time window between irradiation and testing, potential differences in housing, handling, shipping, and other environmental conditions between the two cohorts tested within months of each other. In addition, urine was collected 24 hours after sham-irradiation or irradiation at Brookhaven National Laboratory (BNL), and this additional stressor was not present during the first cohort of 600 mice. As in many genome-wide association studies, we find many markers with small or medium effects. Therefore, we are combining these markers, finding that this increases significance of the results. As more and more genetic markers are added to the analysis, the significance increases and eventually peaks for about 1,000 markers. The identity of these 1,000 markers is not totally overlapping in the two cohorts, although we expect that the structure of the genetic linkage will reveal close relationships between the 1000 markers within each group and also across the marker sets in the two groups of animals. These results are very encouraging. As part of our ongoing analysis, our goal is to distinguish between 1) genetic associations specific to a particular environment or radiation type versus 2) genetic associations that are seen across all environments and radiation types. Our results support that amygdala-dependent phenotypes (i.e., cued fear-related) might be more hard-wired by genetic factors than hippocampus-dependent ones (i.e., contextual fear-related). We measured about 50 different phenotypes but some of them are highly correlated. We consider in our analysis that many phenotypes are highly correlated and likely generated by the same behavioral “factor.” Likewise, many marker associations will be generated by a smaller set of truly causative genes. This approach is complementary to the effort of identifying narrow intervals containing a smaller number of candidate genes.

As a future perspective, we will consider alterations in metabolic products based on a urine metabolomics study of samples collected 24 hours after irradiation. Such potential study might be interesting as it could provide potential noninvasive biomarkers of space radiation-induced cognitive changes. At 800 days post irradiation for each cohort, we returned to CSU (Colorado State University) to dissect the brain. This allows us to follow up on the studies described above with more molecular studies. Our data show that we can find predictive genetic markers and therefore a procedure for classifying naïve mice is feasible. Based on the data set of 600 mice and using genotype information to classify the lowest and the highest phenotype based on cued freezing (amygdala-dependent memory), we showed good classification power and the ability to genotype a naïve mouse and classify it as either likely to be affected or not likely to be affected by radiation. We are currently finalizing the analysis for the first manuscript with the cognitive data.

As a future perspective, we will consider alterations in metabolic products based on a potential urine metabolomics study. The metabolomics study points will be compared/correlated with both the behavioral endpoints and with the genetic information, using the same flexible distance measure approach coupled with Mantel correlations. Such potential study might be interesting as it could provide potential noninvasive biomarkers of space radiation-induced cognitive changes. We anticipate that this analysis will be part of a separate manuscript.

Pertinent to this project in which tumor and behavioral phenotypes are being analyzed in the same mice, a medical student in the Raber Lab (Wendy McGinnis), investigates the effects of immunotherapy and radiation treatment alone and combined in both healthy and tumor-bearing mice using behavioral and cognitive testing and molecular outcome measures. For this project, as part of an active collaboration with Dr. Gough at Providence, a lung carcinoma model in C57BL/6 mice is treated with immunotherapy and peripheral hypofractionated radiation therapy. Based on preliminary studies, the prediction is that combined radiation treatment and immunotherapy will have effects on multiple measures of behavior and cognition, including measures of anxiety, fatigue, and spatial learning and memory.

Major Milestones (Sept. 2014 – Nov. 2015)

	* Completed tumor histopathology (H&E) and compiled data * Began sub-classification of leukemias and lymphomas * Modified DOQTL software to accommodate tumor and ocular phenotypes * Compiled cataract data and started cataract QTL analysis. Oral Presentations Raber, J., Belknap, J., Kleiman, N., Ray, A., Fallgren, C., Kamstock, D., Edmondson, E., Sadowski, A., King, A., Schmidt, D., Weil, M. Characterization of tumor spectrum, cognitive and ocular deficits arising in HZE ion irradiated outbred mice. Radiation Research Annual Meeting 2014, Las Vegas, Nevada, U.S.A Weil, M., Raber, J., Iancu, O., Kleiman, N, Hall, E., Ray, A., Fallgren, C., Kamstock, D., Schmidt, C., King, A., Edmondson, E. Characterization of the tumor spectrum arising in HZE ion irradiated outbred mice. 15th International Congress of Radiation Research (ICRR), 2015, Kyoto, Japan. Weil, M., Raber, J., Iancu, O., Kleiman, N, Hall, E., Ray, A., Fallgren, C., Kamstock, D., Schmidt, C., King, A., Edmondson, E. Characterization of the tumor spectrum arising in HZE ion irradiated outbred mice. Space Radiation and Heavy Ions in Therapy Symposium, 2015, Osaka, Japan. Weil, M. Characterization of the Tumor Spectrum Arising in HZE Ion Irradiated Outbred Mice. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX Kleiman, N.J., Through the Eyes of Astronauts, Cardiologists, Fishermen and Mice: What the eyes can tell us about radiation exposure, genetic determinants of sensitivity and eye disease. LIROT, The Israeli Research Association for Eye Health and Blindness Prevention. Tel Aviv, Israel, 2015 Post-doctoral students supported Adam King, DVM (completed M.S. in Clinical Science) ; Elijah Edmondson, DVM

Bibliography Type:	Description: (Last Updated: 04/27/2026)
Abstracts for Journals and Proceedings	Weil M, Raber J, Iancu O, Kleiman N, Hall E, Ray A, Fallgren C, Kamstock D, Schmidt C, King A, Edmondson E. "Characterization of the tumor spectrum arising in HZE ion irradiated outbred mice." 15th International Congress of Radiation Research (ICRR), Kyoto, Japan, May 25-29, 2015. 15th International Congress of Radiation Research (ICRR), Kyoto, Japan, May 25-29, 2015. , May-2015
Abstracts for Journals and Proceedings	Weil M, Raber J, Iancu O, Kleiman N, Hall E, Ray A, Fallgren C, Kamstock D, Schmidt C, King A, Edmondson E. "Characterization of the tumor spectrum arising in HZE ion irradiated outbred mice." Space Radiation and Heavy Ions in Therapy Symposium 2015, Osaka, Japan, May 22-24, 2015. Space Radiation and Heavy Ions in Therapy Symposium 2015, Osaka, Japan, May 22-24, 2015. , May-2015
Abstracts for Journals and Proceedings	Weil M. "Characterization of the Tumor Spectrum Arising in HZE Ion Irradiated Outbred Mice." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Articles in Peer-reviewed Journals	Barcellos-Hoff MH, Blakely EA, Burma S, Fornace AJ Jr, Gerson S, Hlatky L, Kirsch DG, Luderer U, Shay J, Wang Y, Weil MM. "Concepts and challenges in cancer risk prediction for the space radiation environment." Life Sci Space Res. 2016 Jul;6:82-103. http://dx.doi.org/10.1016/j.lssr.2015.07.006 , Jul-2016
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Articles in Peer-reviewed Journals	Kleiman NJ, Edmondson EF, Weil MM, Fallgren CM, King A, Schmidt C, Hall EJ. "Radiation cataract in heterogeneous stock mice after ?-ray or HZE ion exposure." Life Sci Space Res. 2023 Sep 15. https://doi.org/10.1016/j.lssr.2023.09.004 , Sep-2023
Articles in Peer-reviewed Journals	Chernyavskiy P, Edmondson EF, Weil MM, Little MP. "High-energy particle beam and gamma radiation exposure, familial relatedness and cancer in mice." British Journal of Cancer. 2017 Jun;117(1):41-50. https://doi.org/10.1038/bjc.2017.141 ; PMID: 28535153; PMCID: PMC5520205 , Jun-2017